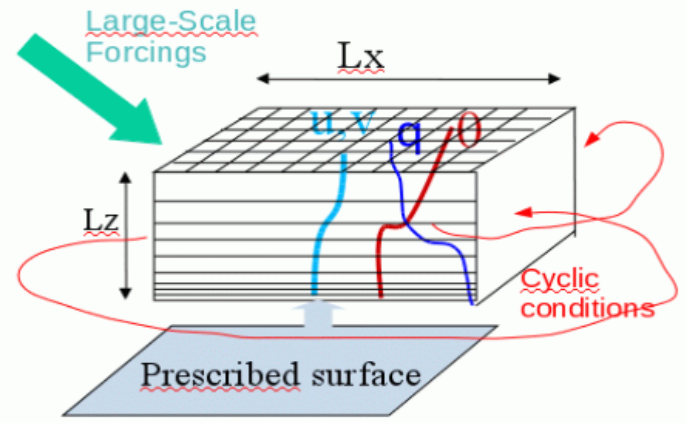
Meso-NH environment
&
Experiments design

MesoNH Tutorial - 2-5 November 2026

Simulation frameworks

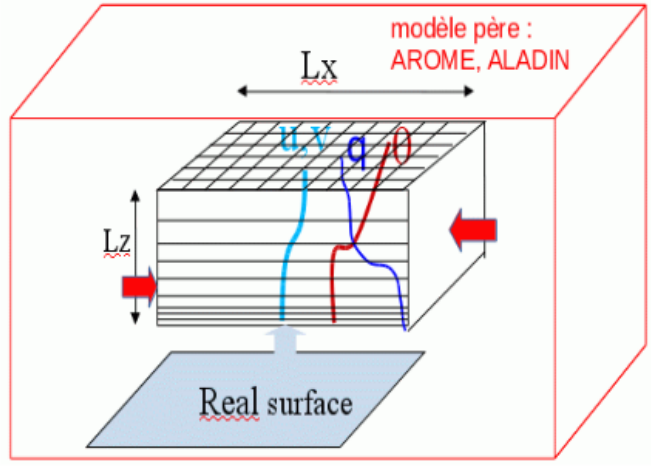
Idealized case



Input
- U, V, T, HU initial profile
- Ug, Vg, T, HU forcing profiles

⇒ initial and forcing horizontally **homogeneous**

Real case



Input
Initial and lateral forcings from a Large-Scale model ⇒ horizontally **heterogeneous**

MESONH simulation = succession of elementary steps

Elementary steps :

1. Production of 2D physiographic file (PGD)
   - PREP_PGD
   - PREP_NEST_PGD
2. Production of 3D init and coupling files
   - PREP_IDEAL_CASE
   - PREP_REAL_CASE
   - SPAWNING
3. Forecast run
   - MESONH
4. Post-processing
   - DIAG
   - SPECTRE

IDEAL CASE

REAL CASE

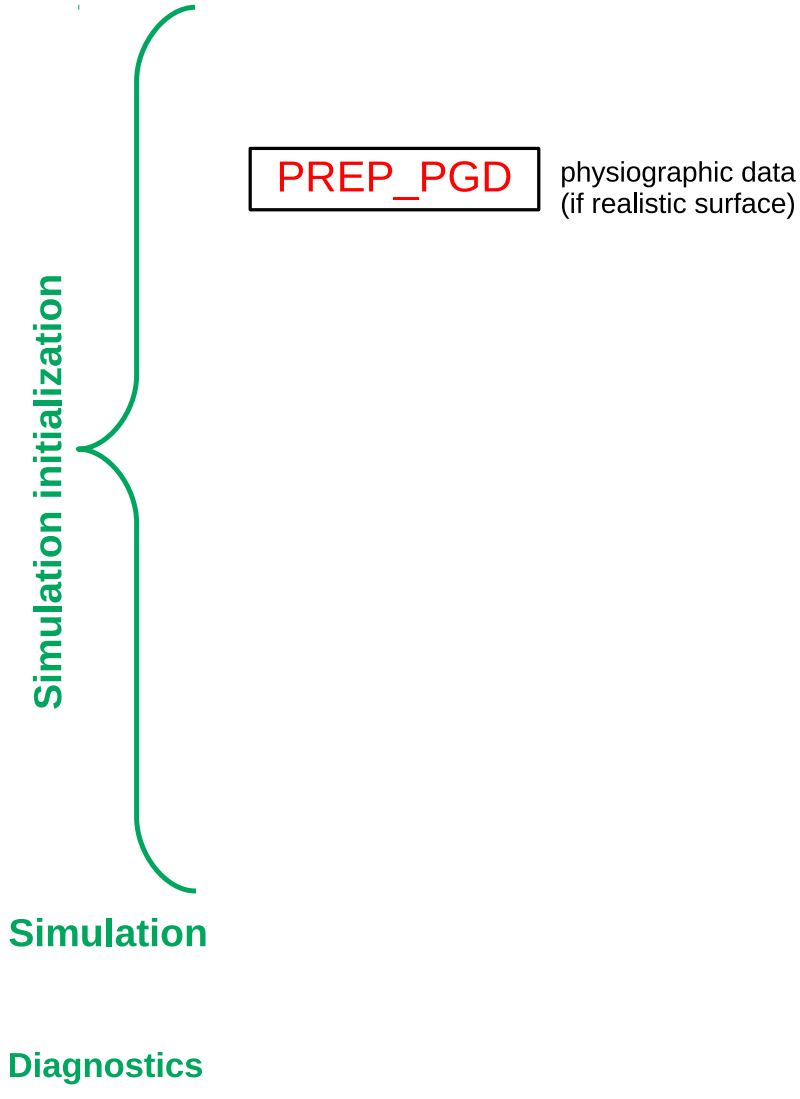


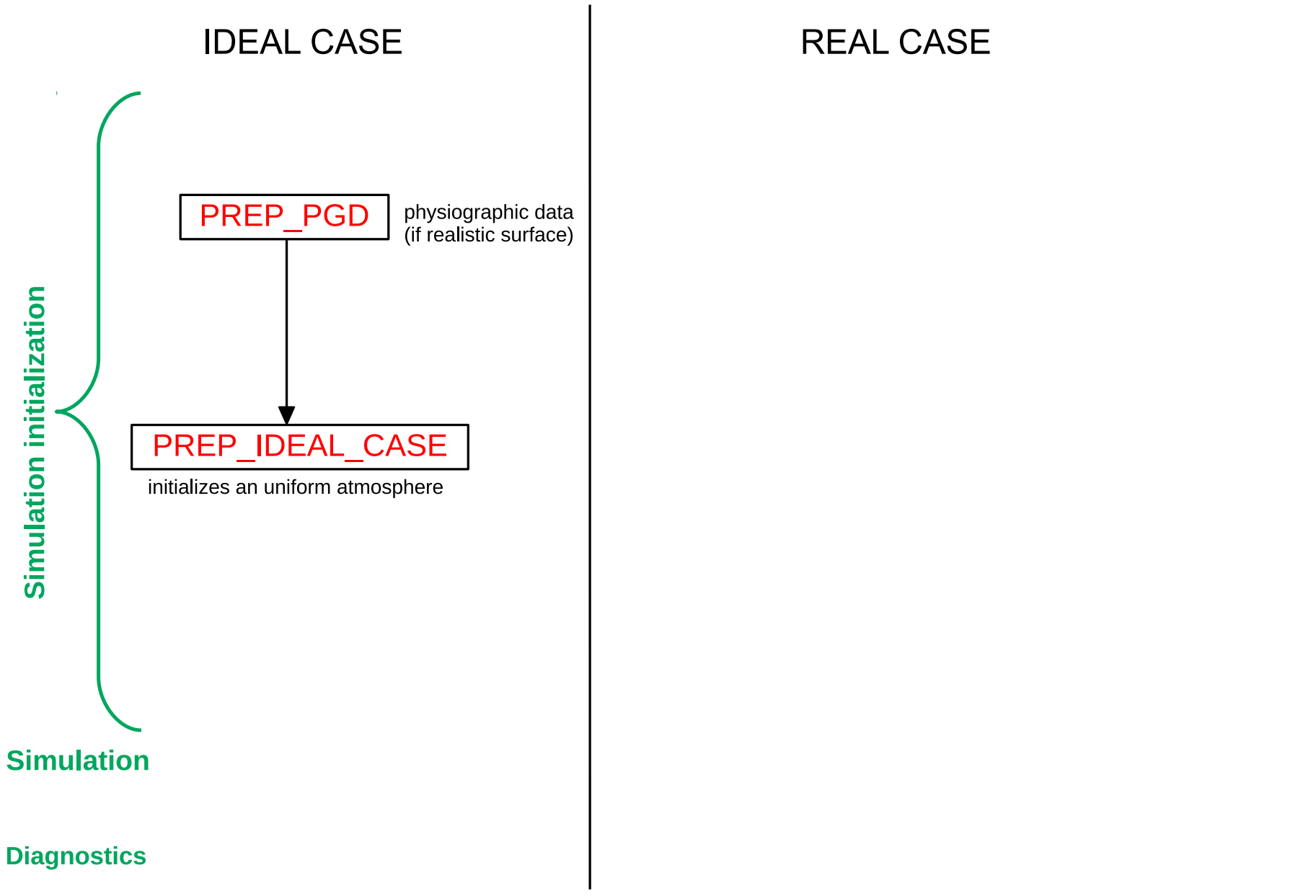
Simulation

Diagnostics

IDEAL CASE

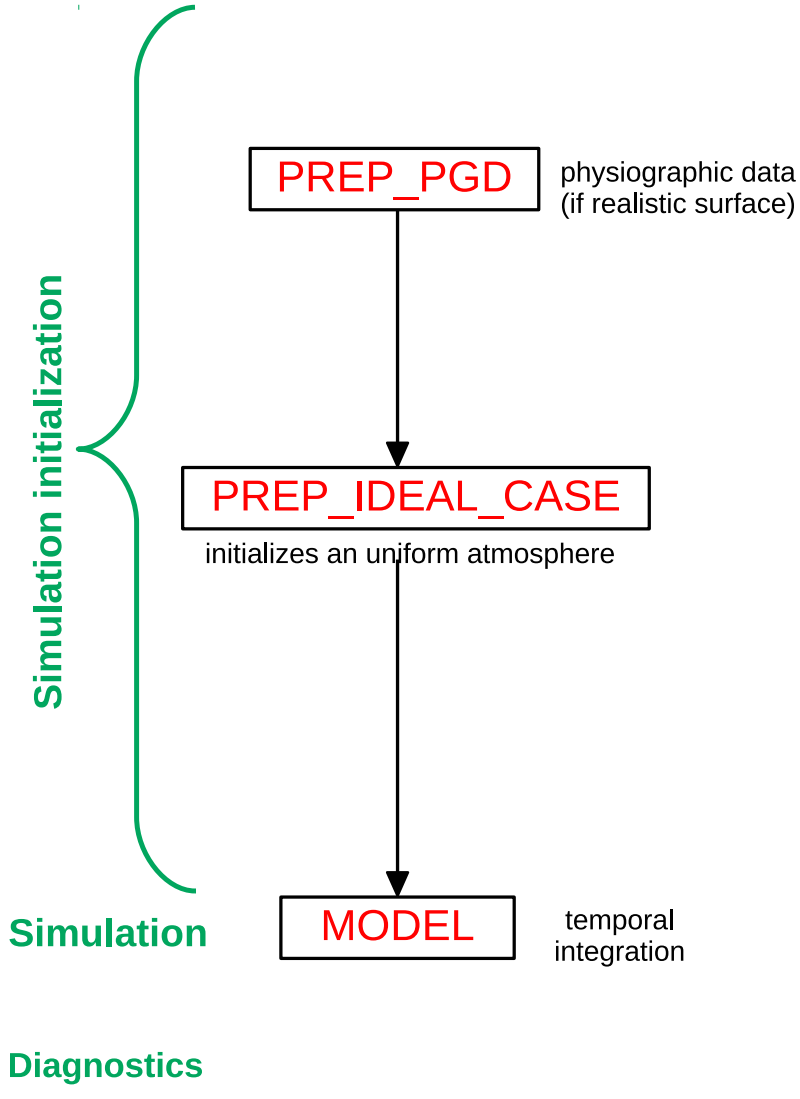
REAL CASE




IDEAL CASE
REAL CASE
PREP_PGD
physiographic data
(if realistic surface)
PREP_IDEAL_CASE
initializes an uniform atmosphere
Simulation initialization
Simulation
Diagnostics

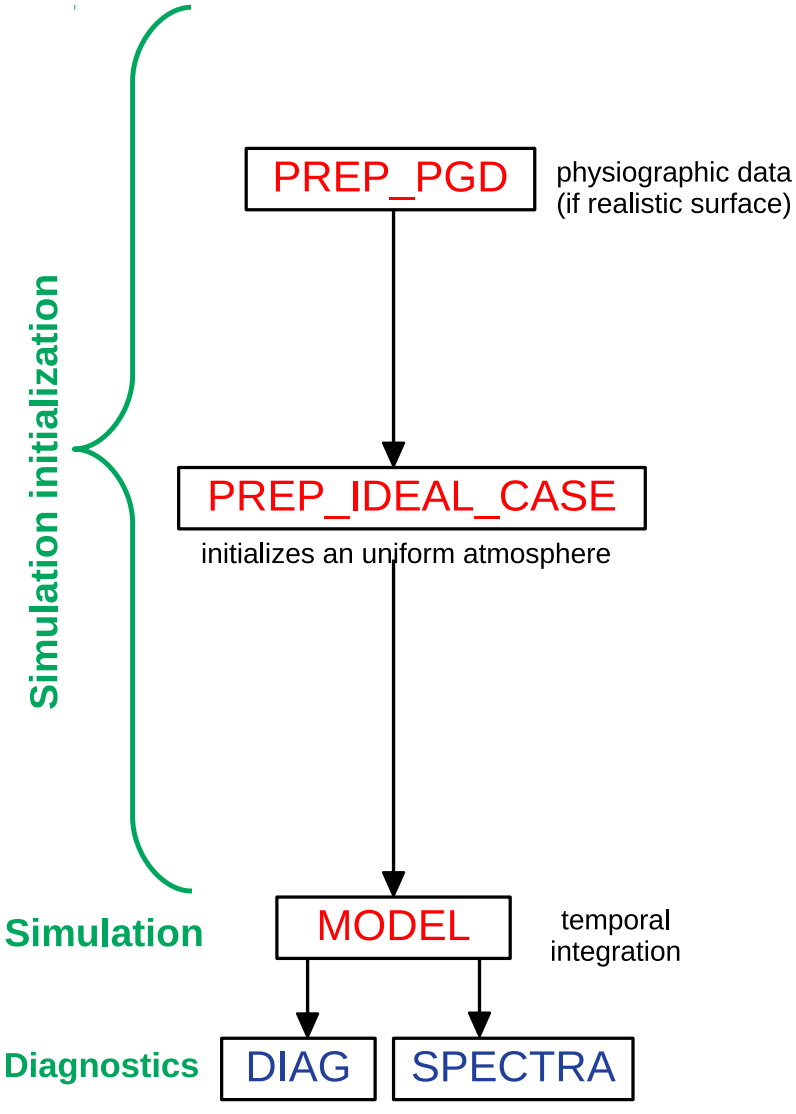
IDEAL CASE

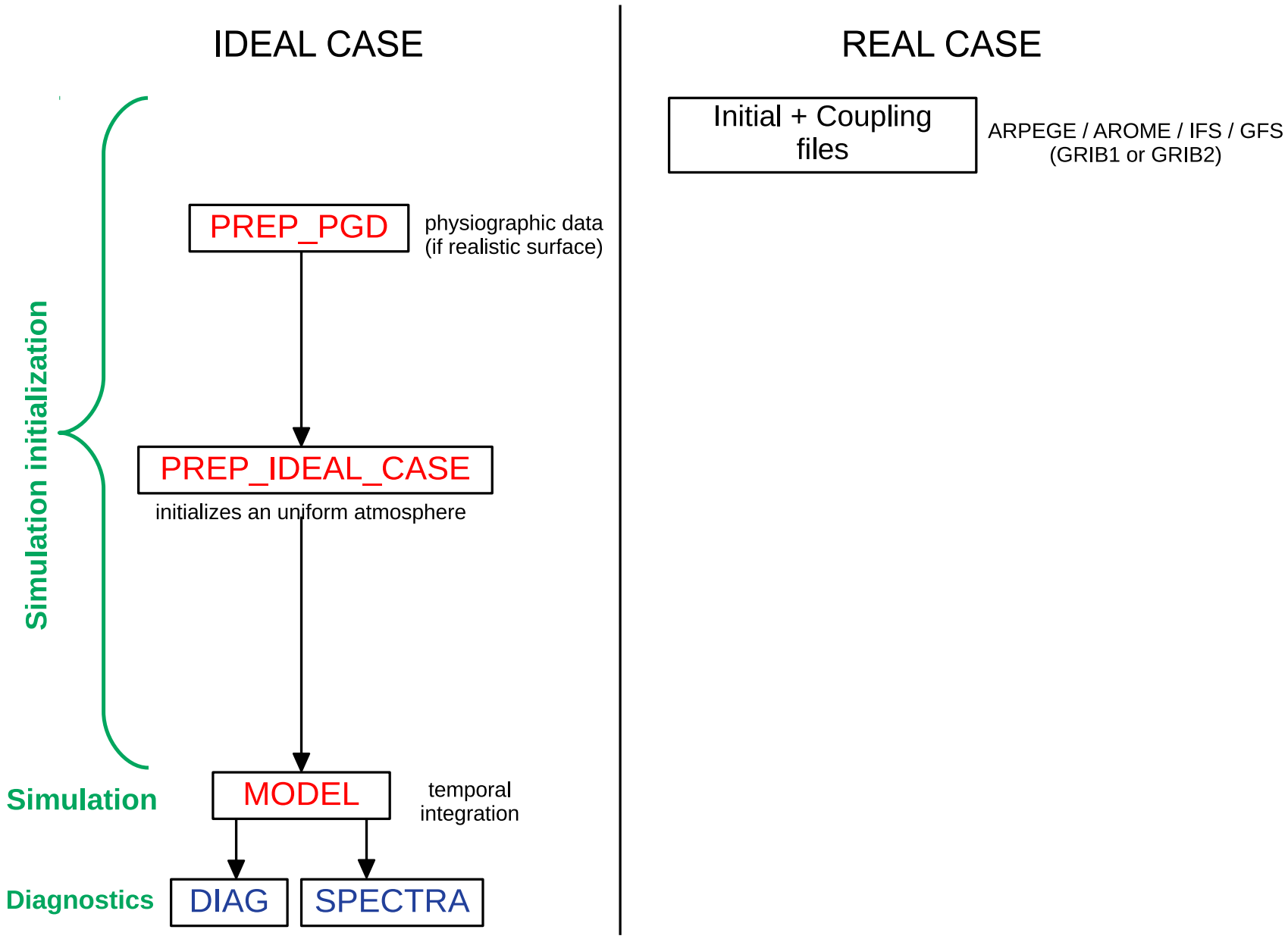
REAL CASE

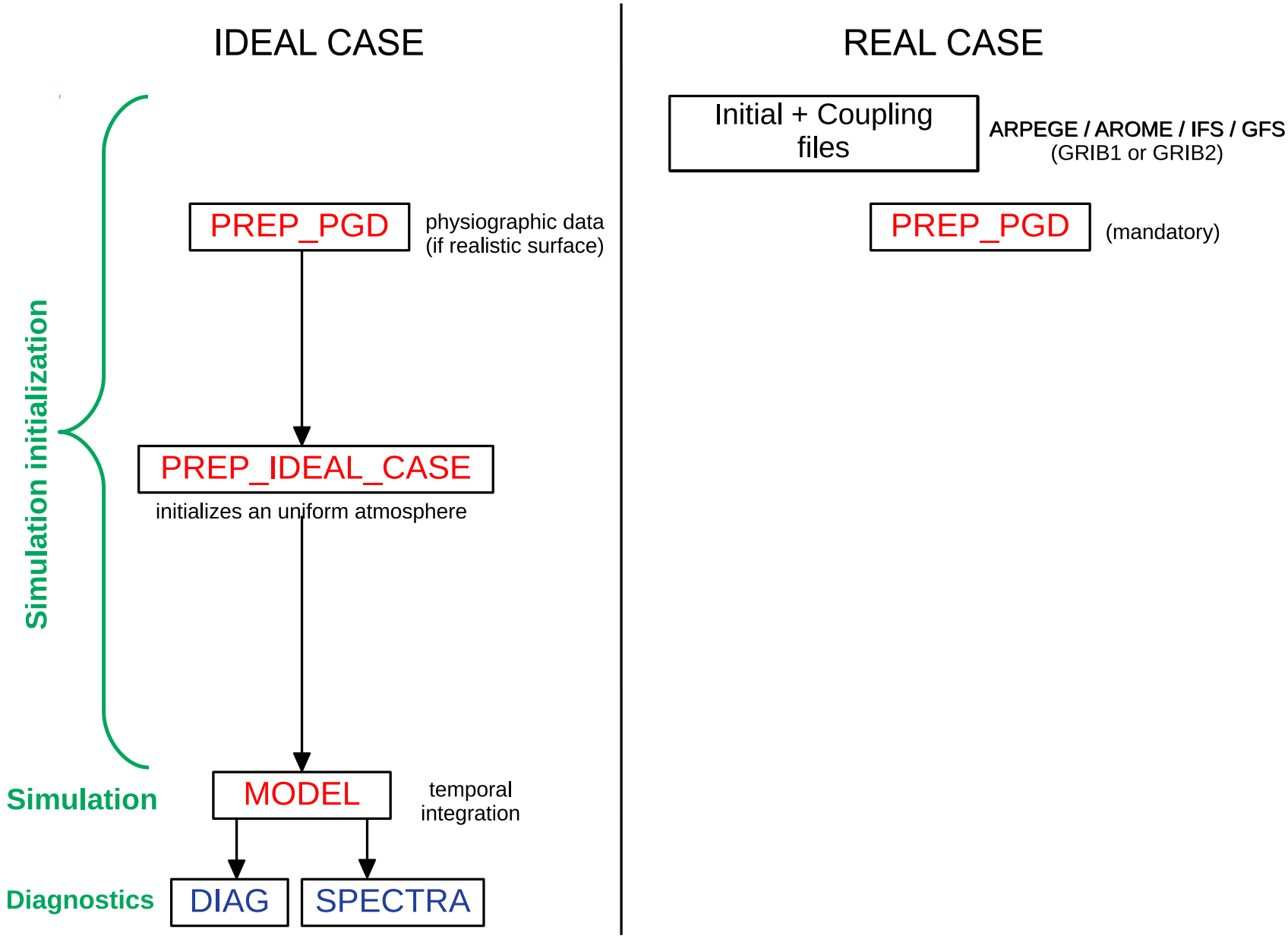


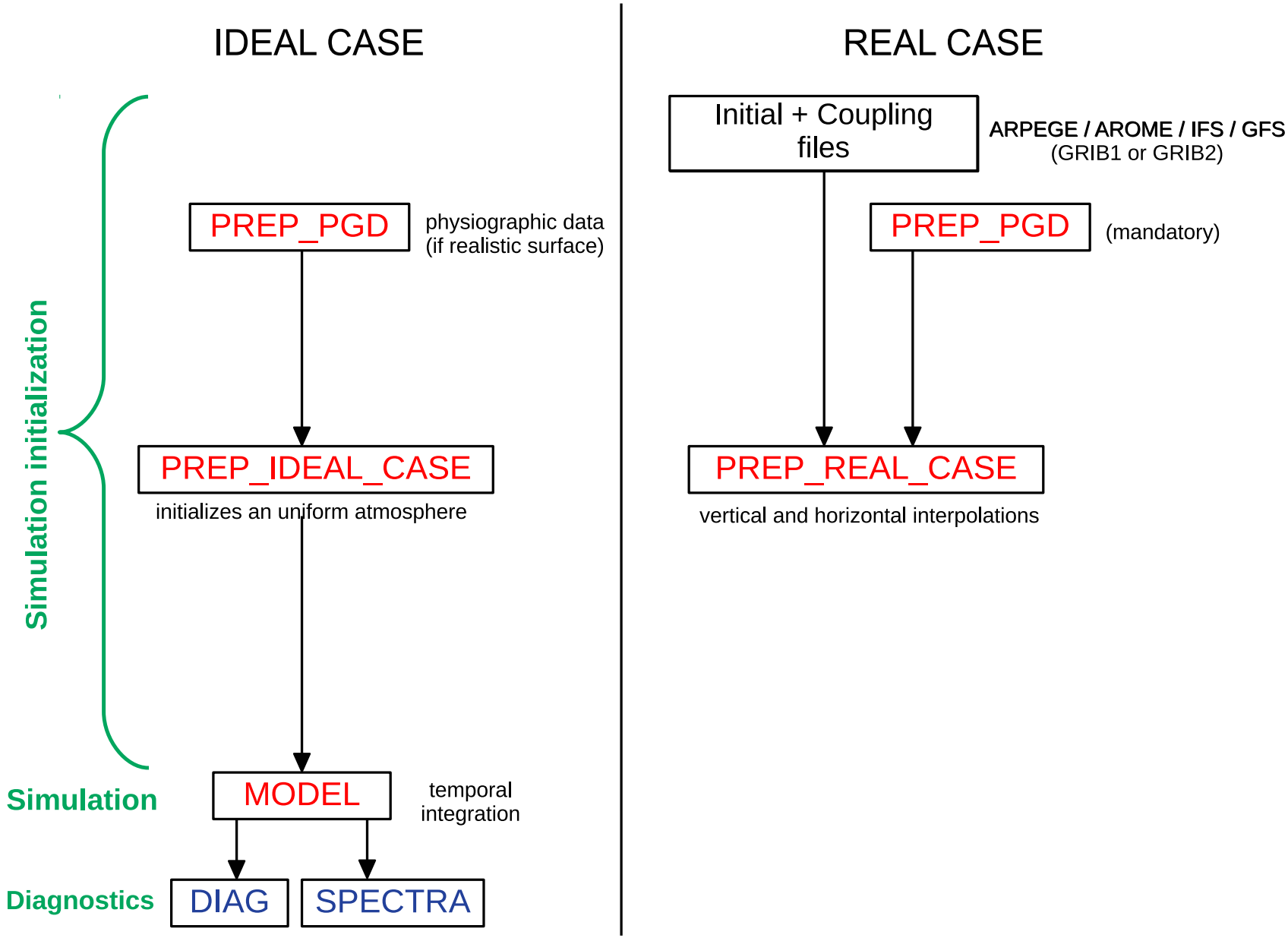
IDEAL CASE

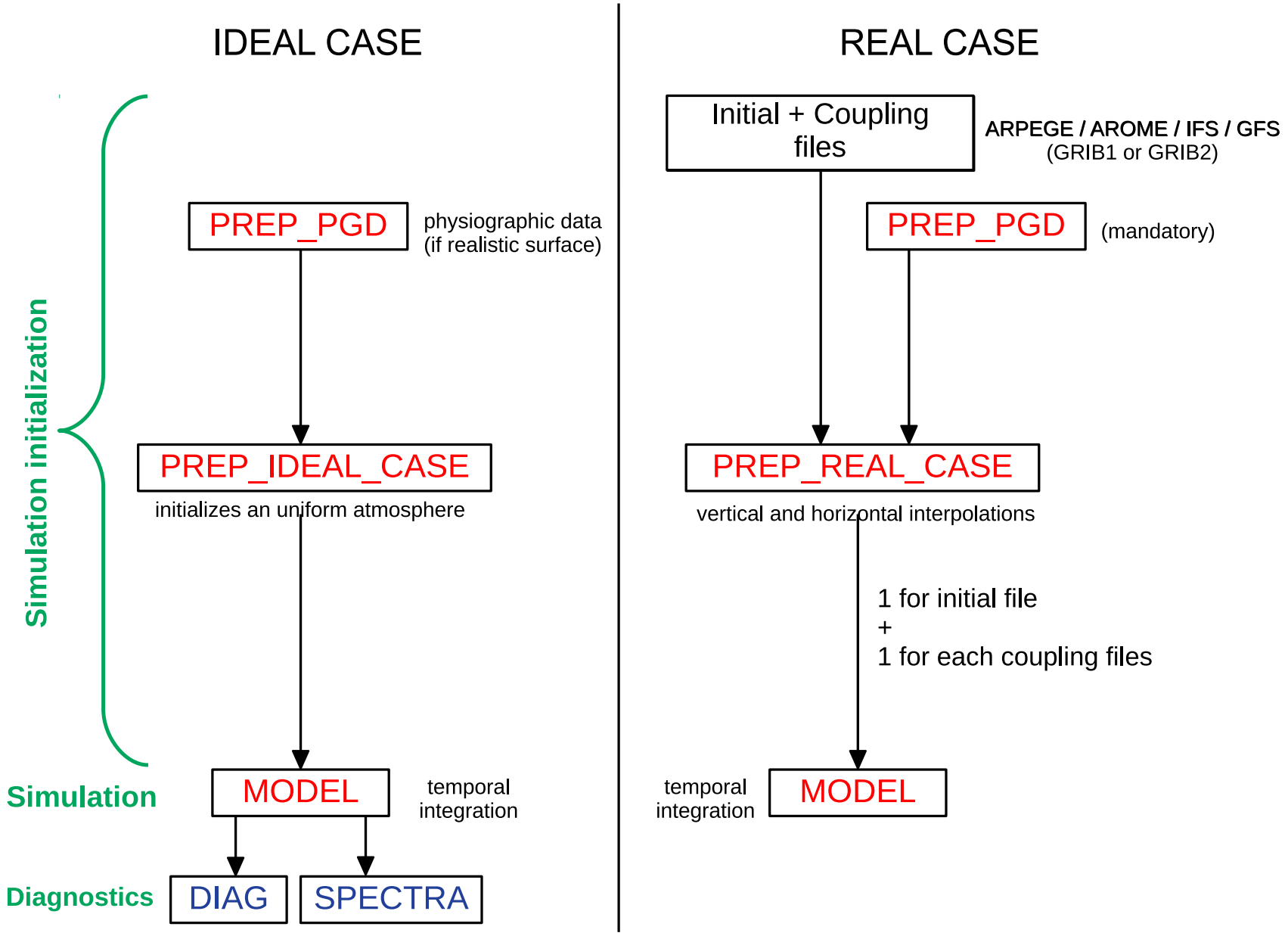
REAL CASE

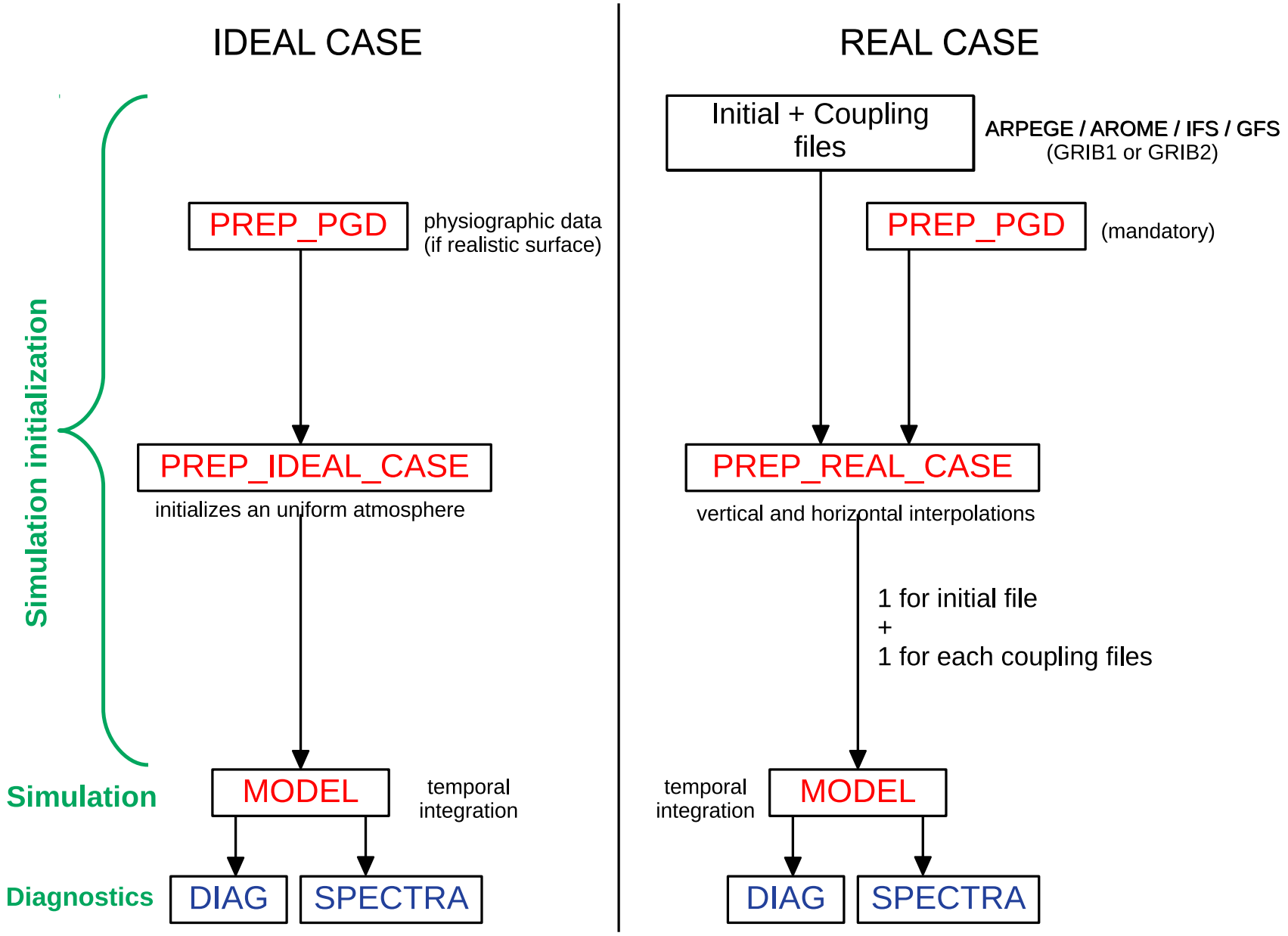



IDEAL CASE
REAL CASE
Initial + Coupling files
ARPEGE / AROME / IFS / GFS
(GRIB1 or GRIB2)
Simulation initialization
PREP_PGD
physiographic data
(if realistic surface)
PREP_IDEAL_CASE
initializes an uniform atmosphere
Simulation
MODEL
temporal
integration
Diagnostics
DIAG
SPECTRA

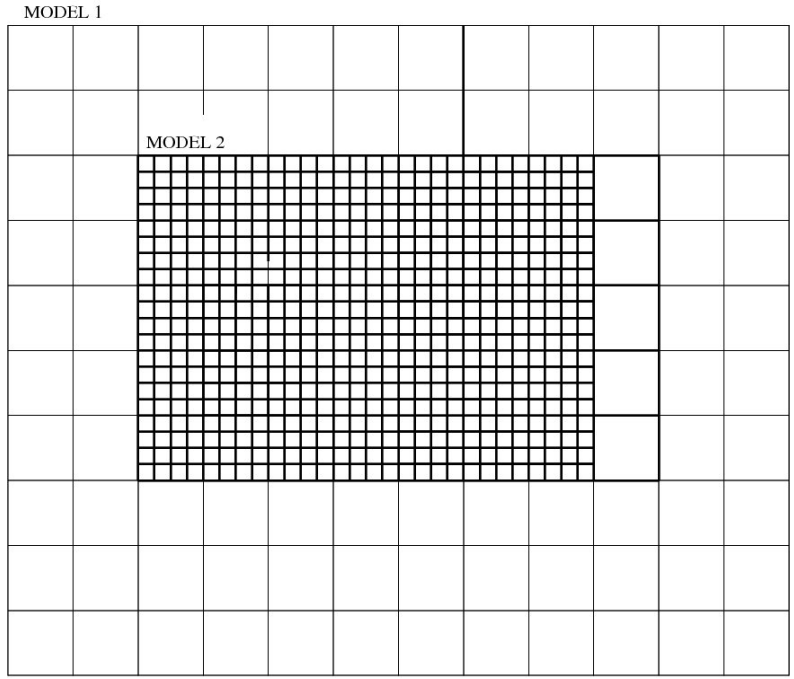

IDEAL CASE
REAL CASE
Initial + Coupling files
ARPEGE / AROME / IFS / GFS
(GRIB1 or GRIB2)
PREP_PGD
physiographic data
(if realistic surface)
PREP_PGD
(mandatory)
Simulation initialization
PREP_IDEAL_CASE
initializes an uniform atmosphere
Simulation
MODEL
temporal
integration
Diagnostics
DIAG
SPECTRA


IDEAL CASE
REAL CASE
Simulation initialization
PREP_PGD
physiographic data
(if realistic surface)
PREP_IDEAL_CASE
initializes an uniform atmosphere
Simulation
MODEL
temporal
integration
Diagnostics
DIAG
SPECTRA
Initial + Coupling
files
ARPEGE / AROME / IFS / GFS
(GRIB1 or GRIB2)
PREP_PGD
(mandatory)
PREP_REAL_CASE
vertical and horizontal interpolations

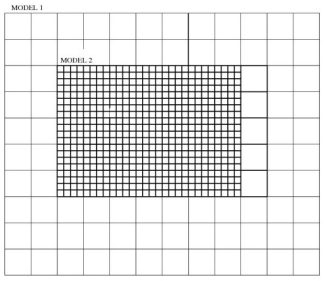

IDEAL CASE
REAL CASE
Simulation initialization
PREP_PGD
physiographic data
(if realistic surface)
PREP_IDEAL_CASE
initializes an uniform atmosphere
Simulation
MODEL
temporal
integration
Diagnostics
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SPECTRA
Initial + Coupling
files
ARPEGE / AROME / IFS / GFS
(GRIB1 or GRIB2)
PREP_PGD
(mandatory)
PREP_REAL_CASE
vertical and horizontal interpolations
1 for initial file
+
1 for each coupling files
temporal
integration
MODEL


IDEAL CASE
REAL CASE
Simulation initialization
PREP_PGD
physiographic data
(if realistic surface)
PREP_IDEAL_CASE
initializes an uniform atmosphere
Simulation
MODEL
temporal
integration
Diagnostics
DIAG
SPECTRA
Initial + Coupling
files
ARPEGE / AROME / IFS / GFS
(GRIB1 or GRIB2)
PREP_PGD
(mandatory)
PREP_REAL_CASE
vertical and horizontal interpolations
1 for initial file
+
1 for each coupling files
temporal
integration
MODEL
DIAG
SPECTRA

two-domain simulation



IDEAL CASE



Flat surface only

Simulation initialization

PREP_IDEAL_CASE

initializes an uniform atmosphere

SPAWNING

horizontal interpolation
mod 1 to mod2

Simulation

MODEL

temporal
integration

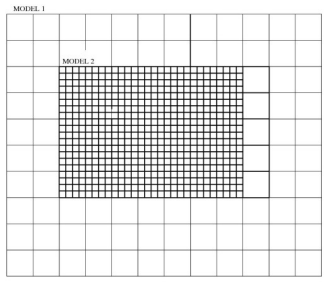
Diagnostics

DIAG

SPECTRA

REAL CASE

IDEAL CASE



Flat surface only

Simulation initialization

PREP_IDEAL_CASE

initializes an uniform atmosphere

SPAWNING

horizontal interpolation
mod 1 to mod2

Simulation

MODEL

temporal
integration

Diagnostics

DIAG

SPECTRA

REAL CASE

Initial + Coupling files

ARPEGE / AROME / IFS / GFS
(GRIB1 or GRIB2)

PREP_PGD

PREP_REAL_CASE

vertical and horizontal interpolations

1 for initial file
+
1 for each coupling files

master domain

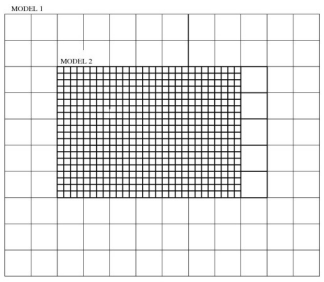
temporal
integration

MODEL

DIAG

SPECTRA

IDEAL CASE



Flat surface only

Simulation initialization

PREP_IDEAL_CASE

initializes an uniform atmosphere

SPAWNING

horizontal interpolation
mod 1 to mod2

Simulation

MODEL

temporal
integration

Diagnostics

DIAG

SPECTRA

REAL CASE

Initial + Coupling files

ARPEGE / AROME / IFS / GFS
(GRIB1 or GRIB2)

PREP_PGD

N times

PREP_NEST_PGD

Orography matching

PREP_REAL_CASE

vertical and horizontal interpolations

1 for initial file
+
1 for each coupling files

master domain

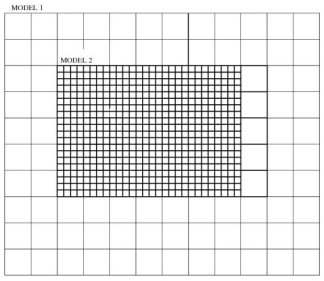
temporal
integration

MODEL

DIAG

SPECTRA

IDEAL CASE



Flat surface only

Simulation initialization

PREP_IDEAL_CASE

initializes an uniform atmosphere

SPAWNING

horizontal interpolation
mod 1 to mod2

Simulation

MODEL

temporal integration

Diagnostics

DIAG

SPECTRA

REAL CASE

Initial + Coupling files

ARPEGE / AROME / IFS / GFS
(GRIB1 or GRIB2)

PREP_PGD

N times

PREP_NEST_PGD

Orography matching

PREP_REAL_CASE

vertical and horizontal interpolations

1 for initial file
+
1 for each coupling files

master domain

N-1 times

SPAWNING

PREP_REAL_CASE

sub-domains

temporal integration

MODEL

DIAG

SPECTRA

IDEAL CASE	REAL CASE
1 domain	1 domain

IDEAL CASE	REAL CASE
1 domain 001_prep_ideal_case 002_run_mesonh	1 domain 12h forecast with 1 coupling file every 3h

IDEAL CASE	REAL CASE
1 domain 001_prep_ideal_case 002_run_mesonh	1 domain 001_prep_pgd 002_prep_real_case x5 (1 init + 4 cpl) 003_run_mesonh 12h forecast with 1 coupling file every 3h
2 domains	2 domains

IDEAL CASE	REAL CASE
1 domain 001_prep_ideal_case 002_run_mesonh	1 domain 001_prep_pgd 002_prep_real_case x5 (1 init + 4 cpl) 003_run_mesonh 12h forecast with 1 coupling file every 3h
2 domains 001_prep_ideal_case 002_spawning_D1_to_D2 003_run_mesonh	2 domains

IDEAL CASE	REAL CASE
1 domain 001_prep_ideal_case 002_run_mesonh	1 domain 001_prep_pgd 002_prep_real_case x5 (1 init + 4 cpl) 003_run_mesonh 12h forecast with 1 coupling file every 3h
2 domains 001_prep_ideal_case 002_spawning_D1_to_D2 003_run_mesonh	2 domains 001_prep_pgd_D1 002_prep_pgd_D2 003_prep_nest_pgd

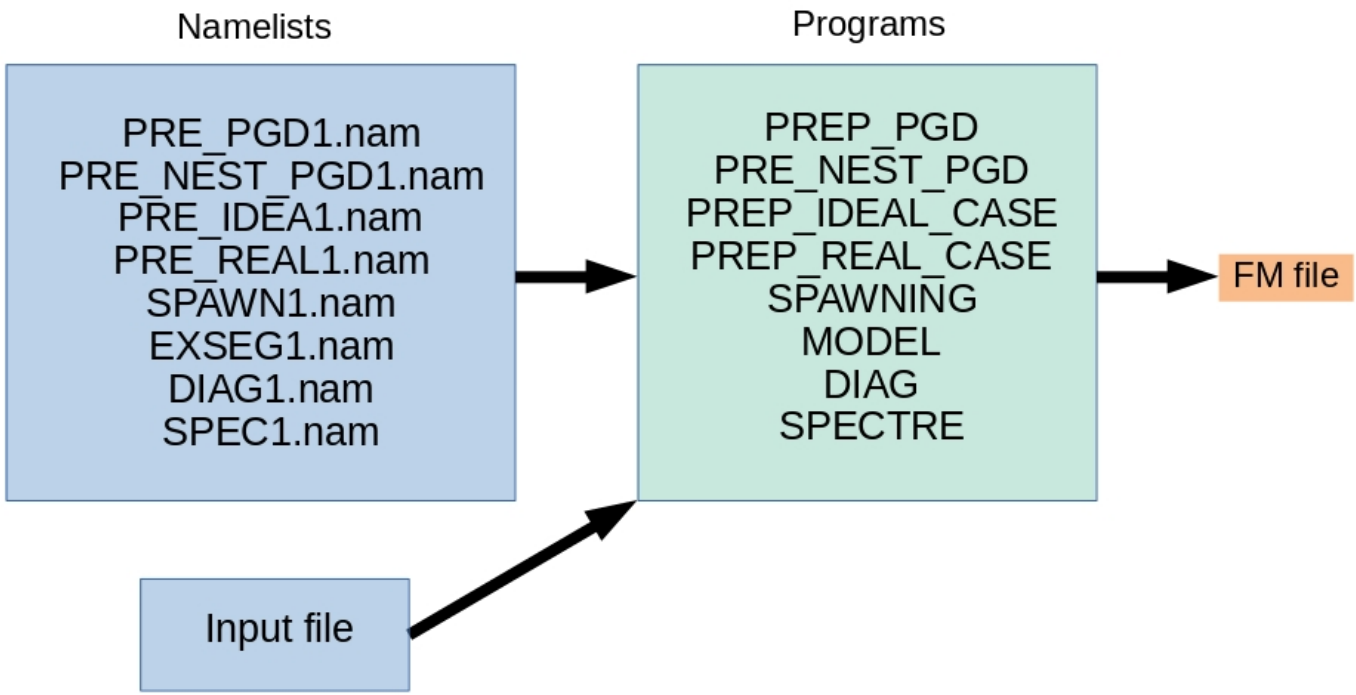
IDEAL CASE	REAL CASE
1 domain 001_prep_ideal_case 002_run_mesonh	1 domain 001_prep_pgd 002_prep_real_case x5 (1 init + 4 cpl) 003_run_mesonh 12h forecast with 1 coupling file every 3h
2 domains 001_prep_ideal_case 002_spawning_D1_to_D2 003_run_mesonh	2 domains 001_prep_pgd_D1 002_prep_pgd_D2 003_prep_nest_pgd 004_prep_real_case_D1 x5 (1 init + 4 cpl) 005_spawning_D1_to_D2 006_prep_real_case_D2 007_run_mesonh

IDEAL CASE	REAL CASE
1 domain 001_prep_ideal_case 002_run_mesonh	1 domain 001_prep_pgd 002_prep_real_case x5 (1 init + 4 cpl) 003_run_mesonh 12h forecast with 1 coupling file every 3h
2 domains 001_prep_ideal_case 002_spawning_D1_to_D2 003_run_mesonh	2 domains 001_prep_pgd_D1 002_prep_pgd_D2 003_prep_nest_pgd 004_prep_real_case_D1 x5 (1 init + 4 cpl) 005_spawning_D1_to_D2 006_prep_real_case_D2 007_run_mesonh
3 domains 001_prep_ideal_case 002_spawning_D1_to_D2	

IDEAL CASE	REAL CASE
1 domain 001_prep_ideal_case 002_run_mesonh	1 domain 001_prep_pgd 002_prep_real_case x5 (1 init + 4 cpl) 003_run_mesonh 12h forecast with 1 coupling file every 3h
2 domains 001_prep_ideal_case 002_spawning_D1_to_D2 003_run_mesonh	2 domains 001_prep_pgd_D1 002_prep_pgd_D2 003_prep_nest_pgd 004_prep_real_case_D1 x5 (1 init + 4 cpl) 005_spawning_D1_to_D2 006_prep_real_case_D2 007_run_mesonh
3 domains 001_prep_ideal_case 002_spawning_D1_to_D2 003_spawning_D2_to_D3 004_run_mesonh	3 domains

IDEAL CASE	REAL CASE
1 domain 001_prep_ideal_case 002_run_mesonh	1 domain 001_prep_pgd 002_prep_real_case x5 (1 init + 4 cpl) 003_run_mesonh 12h forecast with 1 coupling file every 3h
2 domains 001_prep_ideal_case 002_spawning_D1_to_D2 003_run_mesonh	2 domains 001_prep_pgd_D1 002_prep_pgd_D2 003_prep_nest_pgd 004_prep_real_case_D1 x5 (1 init + 4 cpl) 005_spawning_D1_to_D2 006_prep_real_case_D2 007_run_mesonh
3 domains 001_prep_ideal_case 002_spawning_D1_to_D2 003_spawning_D2_to_D3 004_run_mesonh	3 domains 001_prep_pgd_D1 002_prep_pgd_D2 003_prep_pgd_D3 004_prep_nest_pgd 005_prep_real_case_D1 x5 (1 init + 4 cpl) 006_spawning_D1_to_D2 007_prep_real_case_D2 008_spawning_D2_to_D3 009_prep_real_case_D3 010_run_mesonh

Program and Namelists



Namelists

Definition

- Input file for each program
- Use : set the parameters of the program
- Specific format

Format (Méso-NH)

- Name is fixed (except the number if grid-nesting)
- Avoid tabulating
- Contains sub-namelists (groups)
- Groups start by **&NAM_** and end with /
- If a group is not mentionned ⇒ default values

Lists of groups and options : **user's guide** (Méso-NH + SURFEX)

Namelists : example

```
&NAM_LUNITn CINIFILE = "PREP_28JAN21H_D1",
            CINIFILEPGD = "PGD_D1.neste1"/
            CCPLFILE(1) = "PREP_29JAN00H_D1"

&NAM_DYNn XTSTEP = 60., CPRESOPT = "ZRESI",
          NITR=8, LHORELAX_UVWTH = T,
          LHORELAX_RV = T, LVE_RELAX = T,
          NRIMX = 5, NRIMY = 5, XRIMKMAX = 0.0083 /

&NAM_ADVn CUVW_ADV_SCHEME = "WENO_K",
          NWENO_ORDER=5 CTEMP_SCHEME='RK53',
          CMET_ADV_SCHEME = "PPM_01" /

&NAM_PARAMn CTURB = "TKEL", CRAD = "ECRA",
            CSCONV = "EDKF", CDCONV = "KAFR",
            CCLOUD = "ICE3"/
```

Meso-NH files

MesoNH files format

NC (Netcdf)

3 types of output files :

- synchronous **backup**
- synchronous on-demand **output**
- **time series**

with 2 parts :

- **.des** : descriptive ascii file (namelists used)
- **.nc** : data + metadata

Backup files

Backup file = Synchronous file

- all the atmosphere + surface variables **at a given time** on the whole domain
- allows communication between the different programs
- domain dimensions and time are identical for all the fields

In the simulation, a synchronous file allows to (re)start the model in several segments (**RESTART**)

Segments

A MESONH simulation can be divided in multiple **SEGMENTS**.

- subdivide jobs (computing time limit $\Rightarrow$ supercomputer)
  - example : Instead of 1 segment of 24 hours, we can do 4 segments of 6 hours
- have a different number of domain in the segments
  - example : we can have a first segment of 6 hours with 1 domain and a second segment of 12 hours with 2 nested-domains

Backup files : NAM_BACKUP

NAM_BACKUP content

Fortran name	Fortran type	Default value
XBAK_TIME	REAL(:,:)	-999.0
NBAK_STEP	INTEGER(:,:)	-999
XBAK_TIME_FREQ	REAL(:)	-999.0
XBAK_TIME_FREQ_FIRST	REAL(:)	-999.0
NBAK_STEP_FREQ	INTEGER(:)	-999
NBAK_STEP_FREQ_FIRST	INTEGER(:)	-999
LBAK_BEG	LOGICAL	.FALSE.
LBAK_END	LOGICAL	.FALSE.
LBAK_REDUCE_FLOAT_PRECISION	LOGICAL(:)	.FALSE.
LBAK_COMPRESS	LOGICAL(:)	.FALSE.
NBAK_COMPRESS_LEVEL	INTEGER(:)	4
CBAK_DIR	CHARACTER(LEN=512)	

- `XBAK_TIME(m,i)` : array of increments in seconds from the beginning of the segment to the instant where the i-th backup is realized by the model m
- `NBAK_STEP(m,i)` : array of increments in timesteps from the beginning of the segment to the instant where the i-th backup is realized by the model m

Time series

Time series

- contains some chosen variables (flux, tendency, mean) stored at **differents times** during simulation in a part of the domain
- activation of "on-line" diagnostics
- file name ends by .000

Available variables (refer to *Diagnostics* presentation)

- Budgets
- LES
- Aircrafts and balloons
- Stations and profilers

Examples of output files from the run

6 hours run with outputs every 2 hours

1 segment (no RESTART)

Synchronous files (backup)

```
CTRL0.1.SEG01.001
CTRL0.1.SEG01.002
CTRL0.1.SEG01.003
```

Time-series file

```
CTRL0.1.SEG01.000
```

3 segments (RESTART)

1st segment

```
CTRL0.1.SEG01.000
CTRL0.1.SEG01.001
```

2nd segment

```
CTRL0.1.SEG02.000
CTRL0.1.SEG02.001
```

3rd segment

```
CTRL0.1.SEG03.000
CTRL0.1.SEG03.001
```

Highly-frequent and smaller output

On demand smaller output files (optional)

- The user **selects** a few variables only $\Rightarrow$ smaller files
- NetCDF **compression**, single precision possible
- Use : huge domain and/or very frequent output (ex : 3D animation)
- **Restart not possible** from these files
- Available variables : same as in backup files
- **Sub-sampling** on a user defined box

On-demand outputs : NAM_OUTPUT

Note

From version 6.1.0, it is possible to define several output series with different times and variables. This option is used in conjunction with `NOUT_NUMSERIES` (from NAM_CONFIO namelist).

NAM_OUTPUT content

Fortran name	Fortran type	Default value
COUT_DIR	CHARACTER(LEN=512)	''
LOUT_FILESPLIT_DISABLE	LOGICAL(:,:)	.FALSE.
COUT_VAR	CHARACTER(LEN=32)(:,:,:)	''
XOUT_TIME	REAL(:,:,:)	-999.0
NOUT_STEP	INTEGER(:,:,:)	-999
XOUT_TIME_FREQ	REAL(:,:)	-999.0
XOUT_TIME_FREQ_FIRST	REAL(:,:)	XOUT_TIME_FREQ
NOUT_STEP_FREQ	INTEGER(:,:)	-999
NOUT_STEP_FREQ_FIRST	INTEGER(:,:)	XOUT_TIME_FREQ
LOUT_BEG	LOGICAL(:)	.FALSE.
LOUT_END	LOGICAL(:)	.FALSE.

On-demand outputs : NAM_OUTPUT

NAM_OUTPUT content

Fortran name	Fortran type	Default value
LOUT_REDUCE_FLOAT_PRECISION	LOGICAL(:,:)	.FALSE.
LOUT_COMPRESS	LOGICAL(:,:)	.FALSE.
COUT_COMPRESS_ALGO	CHARACTER(LEN=10)(:,:)	'ZSTD'
NOUT_COMPRESS_LEVEL	INTEGER(:,:)	4
LOUT_COMPRESS_LOSSY	LOGICAL(:,:)	.FALSE.
COUT_COMPRESS_LOSSY_ALGO	CHARACTER(LEN=10)(:,:)	'GRANULARBR'
NOUT_COMPRESS_LOSSY_NSD	INTEGER(:,:)	3

On-demand outputs : NAM_OUTPUT

NAM_OUTPUT content

Fortran name	Fortran type	Default value
NOUT_VAR_REDUCE_FLOAT_PRECISION	INTEGER(:,:,:)	-888
NOUT_VAR_COMPRESS_LEVEL	INTEGER(:,:,:)	-888
COUT_VAR_COMPRESS_LOSSY_ALGO	CHARACTER(LEN=10)(:,:,:)	'NOT_SET'
NOUT_VAR_COMPRESS_LOSSY_NSD	INTEGER(:,:,:)	-888
XOUT_VAR_THR_MIN	REAL(:,:,:)	not set
XOUT_VAR_THR_MAX	REAL(:,:,:)	not set
XOUT_VAR_THR_ABSMIN	REAL(:,:,:)	not set
XOUT_VAR_THR_ABSMAX	REAL(:,:,:)	not set
COUT_VAR_THR_MIN_BEHAVIOR	CHARACTER(LEN=9)(:,:,:)	'NOT_SET'
COUT_VAR_THR_MAX_BEHAVIOR	CHARACTER(LEN=9)(:,:,:)	'NOT_SET'
COUT_VAR_THR_ABSMIN_BEHAVIOR	CHARACTER(LEN=9)(:,:,:)	'NOT_SET'
COUT_VAR_THR_ABSMAX_BEHAVIOR	CHARACTER(LEN=9)(:,:,:)	'NOT_SET'
XOUT_VAR_RND_FACTOR	REAL(:,:,:)	-999.0

On-demand outputs : NAM_OUTPUT

Fortran name	Fortran type	Default value
NOUT_BOXES	INTEGER(:,:)	0
COUT_BOX_NAME	CHARACTER(LEN=32)(:,:,:)	'Box_nnnn'
NOUT_BOX_IINF	INTEGER(:,:,:)	-999.0
NOUT_BOX_ISUP	INTEGER(:,:,:)	-999.0
NOUT_BOX_JINF	INTEGER(:,:,:)	-999.0
NOUT_BOX_JSUP	INTEGER(:,:,:)	-999.0
NOUT_BOX_KINF	INTEGER(:,:,:)	-999.0
NOUT_BOX_KSUP	INTEGER(:,:,:)	-999.0
COUT_BOX_VAR_SUPP	CHARACTER(LEN=32)(:,:,:,:)	''
LOUT_MAINDOMAIN_WRITE	LOGICAL(:,:)	.FALSE.